WILLIAM DUAN

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EDUCATION

Ph.D. in Economics, University of British Columbia 2020 – 2027 (expected)
Co-supervisors: Hao Li and Vitor Farinha Luz. Committee member: Wei Li.

M.A. in Economics, University of British Columbia 2019 – 2020

B.A., First Class Honours in Economics, McGill University 2015 – 2019
Minor in Mathematics and Computer Science. GPA 3.92.

FIELDS

Primary: Microeconomic Theory
Secondary: Mechanism Design; Information Economics

JOB MARKET PAPER

Selling Mechanisms under Misspecified Learning
Presented at the Canadian Economics Association Annual Meetings, Vancouver, May 2026.

Buyers in a repeated selling mechanism hold a misspecified parametric model of their environment and learn by Bayesian updating from allocation outcomes alone. Because the mechanism generates the outcomes buyers learn from, its design shapes what they come to believe. This paper asks when a seller can choose a mechanism under which a given, possibly misspecified, belief is sustained by learning, and characterizes such mechanisms. With a single buyer, conditions on the buyer's model alone guarantee that a unique mechanism draws every initial belief to the target belief. With two buyers, local stability is established under conditions on the buyers' models and the value distribution, and a computational method identifies mechanisms that extend the basin of attraction to every belief below the sustained one. The main revenue implication is that value-dependent beliefs can allow the seller to exceed the Myerson benchmark, whereas value-independent beliefs cannot.

WORKING PAPERS

Ex-ante Heterogeneity in Directed Search Games
Shows that the standard symmetric equilibrium in directed search with finitely many workers is fragile to small cross-worker differences in skill distributions, and characterizes a robust equilibrium that always exists and connects it to the tractable symmetric benchmark.

FELLOWSHIPS, SCHOLARSHIPS AND AWARDS

Agustín Beltrán Award for Excellence in Teaching, Vancouver School of Economics	2024, 2025, 2026
SSHRC Doctoral Fellowship	2020 – 2024
President's Academic Excellence Initiative PhD Award, UBC	2020 – 2024
A. D. Scott Fellowship in Economics, UBC	2020
Faculty of Arts Graduate Award, UBC	2019 – 2020
Hugh Brock Scholarship, McGill University	2015 – 2018
J. W. Birch Economics Undergraduate Award, McGill University	2017
Dean's Honour List, McGill University	2015 – 2017

TEACHING EXPERIENCE

Teaching Assistant, Vancouver School of Economics, UBC 2019 – 2026

- ECON 600/601 Advanced Microeconomic Theory I/II (Ph.D. core) 2021 – 2026
  Agustín Beltrán Award for Excellence in Teaching, 2024, 2025, 2026.
  Co-designed ECON 601 problem sets, 2025, 2026.
- ECON 101 Principles of Microeconomics 2020, Summer 2025
- ECON 514 Information and Incentives 2021
- ECON 303 Intermediate Microeconomic Analysis II 2020
- ECON 355 Introduction to International Trade 2019

Instructor, VSE Math Camp (M.A. and Ph.D. sessions) Summer 2026

Designed the Ph.D. sessions with Prof. Jesse Perla and wrote their slides and practice problems. For the M.A. sessions, which ran on recorded faculty lectures, led the problem sessions and questions.

RESEARCH EXPERIENCE

Research Assistant to Prof. Vitor Farinha Luz, University of British Columbia 2022

Research Assistant to Prof. Hassan Benchekroun, McGill University 2017 – 2019

Theoretical derivations for dynamic programming models and numerical simulation of models in international trade and resource economics.

SKILLS

Languages: English and Mandarin Chinese (fluent); French (basic)
Software: Python, R, Julia, MATLAB, Mathematica, Java; LaTeX, Quarto, Git

REFERENCES

Prof. Hao Li (co-supervisor)
Vancouver School of Economics, UBC
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Prof. Vitor Farinha Luz (co-supervisor)
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Prof. Wei Li
Vancouver School of Economics, UBC
Wei.Li@ubc.ca

Prof. Jesse Perla
Vancouver School of Economics, UBC
jesse.perla@ubc.ca

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